Practical-6

1. Subquery in MySQL

A **subquery** is a query inside another query. It executes **first**, and its result is used by the main query.

Example 1: Subquery in WHERE clause

Find employees earning more than **average salary**.

```
mysql> INSERT INTO Department VALUES
    -> (1, 'HR'),
    -> (2, 'IT'),
    -> (3, 'Finance');
Query OK, 3 rows affected (0.06 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Employee VALUES
    -> (101, 'Amit', 50000, 2),
    -> (102, 'Ravi', 60000, 2),
    -> (103, 'Neha', 40000, 1),
    -> (104, 'Priya', 70000, 3),
    -> (105, 'Karan', 55000, 2);
Query OK, 5 rows affected (0.04 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Employee;
+--------+-------+--------+---------+
| emp_id | name  | salary | dept_id |
+--------+-------+--------+---------+
|    101 | Amit  |  50000 |       2 |
|    102 | Ravi  |  60000 |       2 |
|    103 | Neha  |  40000 |       1 |
|    104 | Priya |  70000 |       3 |
|    105 | Karan |  55000 |       2 |
+--------+-------+--------+---------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Department;
+---------+-----------+
| dept_id | dept_name |
+---------+-----------+
|       1 | HR        |
|       2 | IT        |
|       3 | Finance   |
+---------+-----------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT AVG(salary) FROM Employee;
+-------------+
| AVG(salary) |
+-------------+
|  55000.0000 |
+-------------+
1 row in set (0.02 sec)

mysql> SELECT name, salary
    -> FROM Employee
    -> WHERE salary > (SELECT AVG(salary) FROM Employee);
+-------+--------+
| name  | salary |
+-------+--------+
| Ravi  |  60000 |
| Priya |  70000 |
+-------+--------+
2 rows in set (0.02 sec)
```

Example 2: Subquery with IN

Find employees working in "IT" department.

```
mysql> SELECT dept_id
    -> FROM Department
    -> WHERE dept_name = 'IT';
+---------+
| dept_id |
+---------+
|       2 |
+---------+
1 row in set (0.01 sec)

mysql> SELECT name
    -> FROM Employee
    -> WHERE dept_id IN (
    ->     SELECT dept_id
    ->     FROM Department
    ->     WHERE dept_name = 'IT'
    -> );
+-------+
| name  |
+-------+
| Amit  |
| Ravi  |
| Karan |
+-------+
3 rows in set (0.01 sec)
```

Example 3: Subquery in FROM clause

```
mysql> SELECT dept_id, salary
    -> FROM Employee;
+---------+--------+
| dept_id | salary |
+---------+--------+
|       2 |  50000 |
|       2 |  60000 |
|       1 |  40000 |
|       3 |  70000 |
|       2 |  55000 |
+---------+--------+
5 rows in set (0.00 sec)

mysql> SELECT dept_id, AVG(salary)
    -> FROM (
    ->     SELECT dept_id, salary
    ->     FROM Employee
    -> ) AS temp
    -> GROUP BY dept_id;
+---------+-------------+
| dept_id | AVG(salary) |
+---------+-------------+
|       1 |  40000.0000 |
|       2 |  55000.0000 |
|       3 |  70000.0000 |
+---------+-------------+
3 rows in set (0.00 sec)
```

2. Correlated Subquery in MySQL

A **correlated subquery** depends on values from the outer query and runs **once for each row**.

Example 1: Salary greater than department average

```
mysql> SELECT e1.name, e1.salary, e1.dept_id
    -> FROM Employee e1
    -> WHERE e1.salary > (
    ->     SELECT AVG(e2.salary)
    ->     FROM Employee e2
    ->     WHERE e2.dept_id = e1.dept_id
    -> );
+------+--------+---------+
| name | salary | dept_id |
+------+--------+---------+
| Ravi |  60000 |       2 |
+------+--------+---------+
1 row in set (0.00 sec)

mysql>
```

Example 2: Find customers with orders

```
mysql> SELECT * FROM CUSTOMER;
+-------------+---------------+
| customer_id | customer_name |
+-------------+---------------+
|           1 | Amit          |
|           2 | Ravi          |
|           3 | Neha          |
+-------------+---------------+
3 rows in set (0.14 sec)
```

```
mysql> SELECT * FROM ORDERS
    -> ;
+----------+-------------+
| order_id | customer_id |
+----------+-------------+
|      101 |           1 |
|      102 |           1 |
|      103 |           2 |
+----------+-------------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT c.customer_name
    -> FROM Customer c
    -> WHERE EXISTS (
    ->     SELECT *
    ->     FROM Orders o
    ->     WHERE o.customer_id = c.customer_id
    -> );
+---------------+
| customer_name |
+---------------+
| Amit          |
| Ravi          |
+---------------+
2 rows in set (0.28 sec)
```

Example 3: Maximum salary in each department

```
mysql> SELECT e1.*
    -> FROM Employee e1
    -> WHERE salary = (
    ->     SELECT MAX(e2.salary)
    ->     FROM Employee e2
    ->     WHERE e2.dept_id = e1.dept_id
    -> );
+--------+-------+--------+---------+
| emp_id | name  | salary | dept_id |
+--------+-------+--------+---------+
|    102 | Ravi  |  60000 |       2 |
|    103 | Neha  |  40000 |       1 |
|    104 | Priya |  70000 |       3 |
+--------+-------+--------+---------+
3 rows in set (0.20 sec)
```

Difference Between Subquery and Correlated Subquery

Subquery	Correlated Subquery
Executes once	Executes for each row
Independent	Depends on outer query
Faster	Slower
Simple logic	Complex filtering